In [4]:
```
# Importing Libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import librosa  # Used for audio analysis
import librosa.display
from glob import glob # Used for file handling
import IPython.display as ipd # Interactive audio playback

# Libraries used for analysis
import statsmodels.api as sm
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
```

In [5]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

In [6]:
```
# loading the dataset

audio_dir=glob("/content/drive/MyDrive/Hindi_fem_mono/Hindi_fem_audio/*.wav")
audio_file = sorted(audio_dir)
```

In [7]:
```
# play a file

ipd.Audio(audio_file[2])
```

Out[7]: Your browser does not support the audio element.

Basic EDA

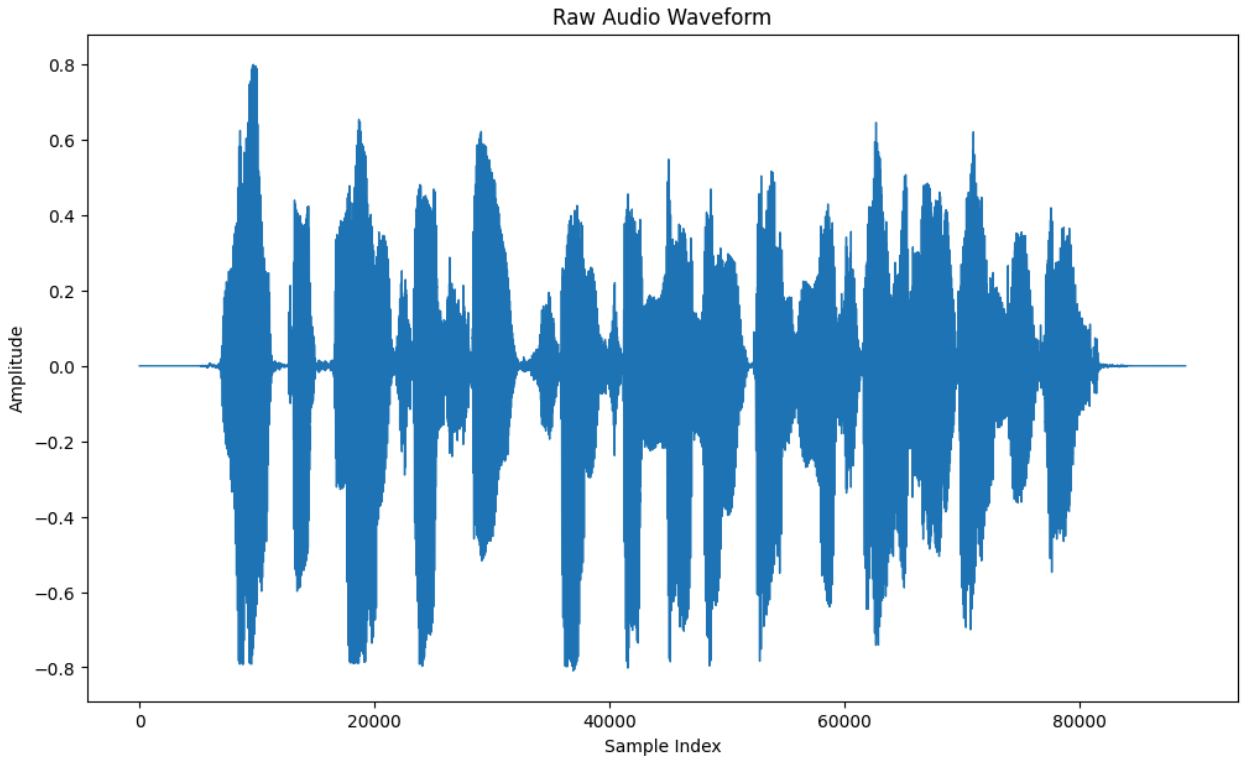
In [8]:
```
# load the audio cell file
# y will contain the raw audio time series data
# sr will be the sample rate

y,sr = librosa.load(audio_file[2])
```

In [9]:
```
# display the first 10 samples of the raw audio data
print(f'the first 10 samples of y is {y[:10]}')
# Display the shape of the audio data array
print(f"Shape of y (audio data): {y.shape}")

# Display the sample rate of the audio
print(f"Sample rate: {sr} Hz")
```

```
the first 10 samples of y is [-1.4012909e-05  8.9582920e-07 -9.1765742e-06 -2.4
880851e-06
  1.5019210e-05  1.0491634e-05 -4.4989625e-05 -5.4112879e-06
 -1.2322516e-06  3.0269127e-06]
Shape of y (audio data): (89011,)
Sample rate: 22050 Hz
```

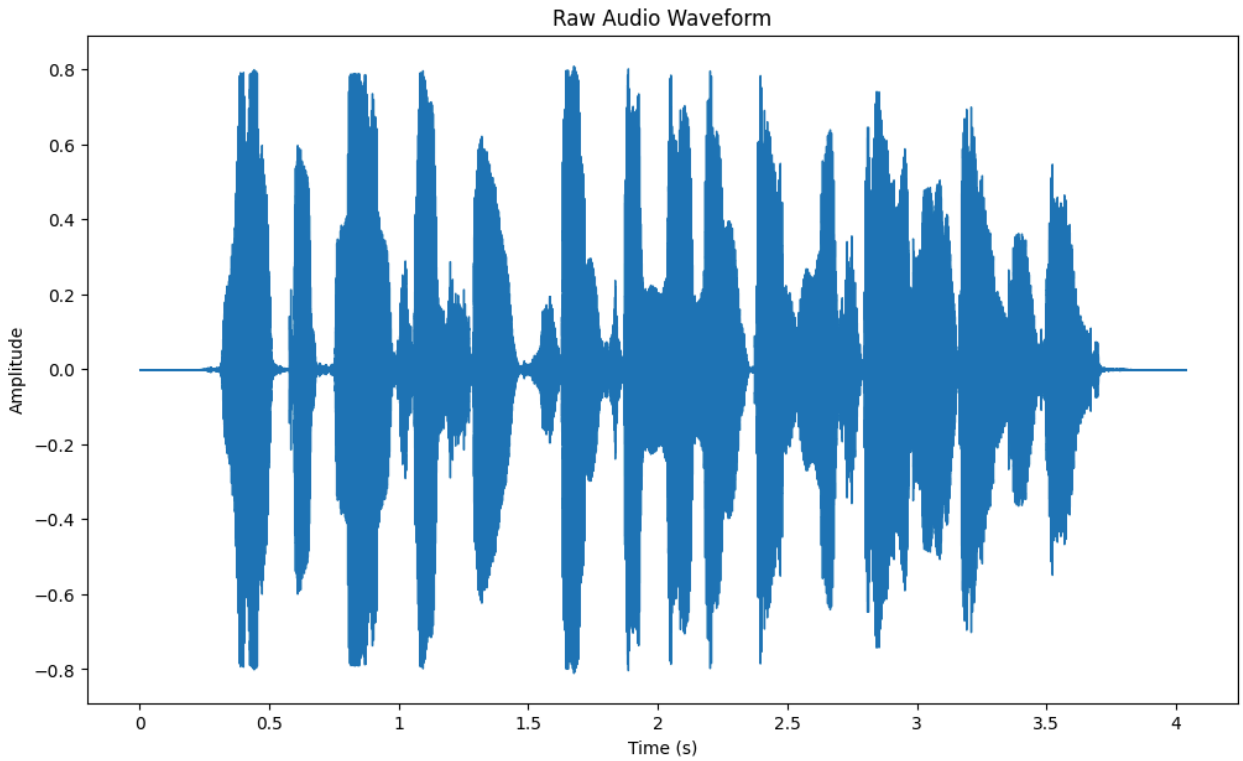
In [10]:

```
# Plot the raw audio waveform
pd.Series(y).plot(
    figsize=(12, 7),             # Set the figure size for better visibility
    lw=1,                       # Line width for the waveform
    title="Raw Audio Waveform", # Title of the plot        # Set the color usi
    xlabel="Sample Index",      # Label for the x-axis
    ylabel="Amplitude"          # Label for the y-axis
)
plt.show()
```



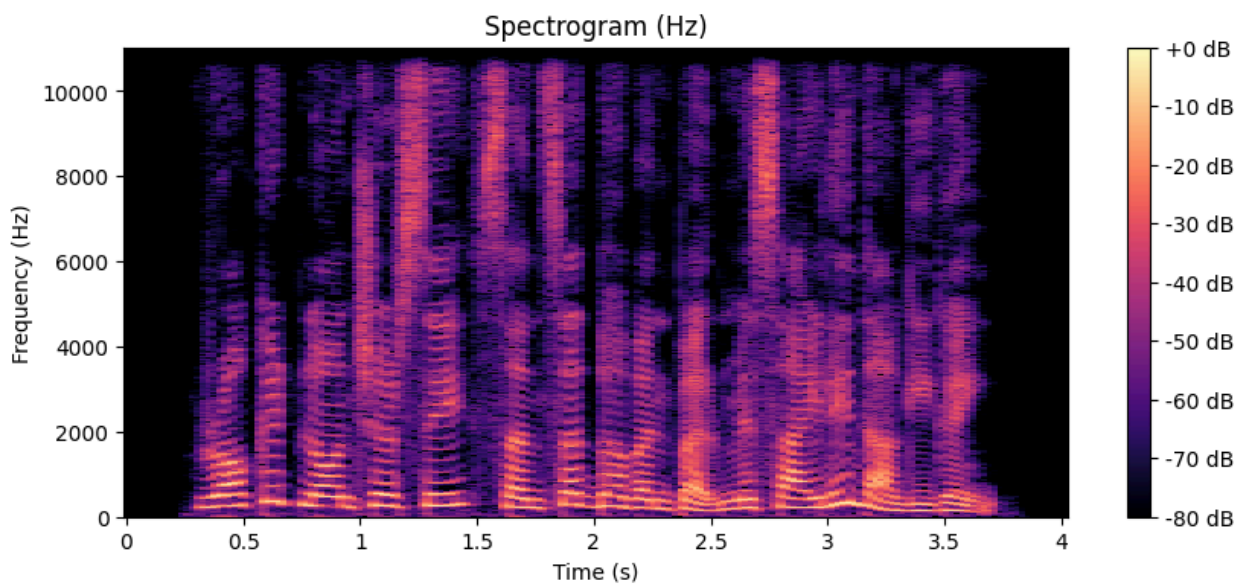
In [11]:

```
plt.figure(figsize=(12,7))
# Reload y and sr to ensure they refer to the audio data for plotting
y, sr = librosa.load(audio_file[2])
librosa.display.waveshow(y=y, sr=sr)
plt.title('Raw Audio Waveform')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.show()
```



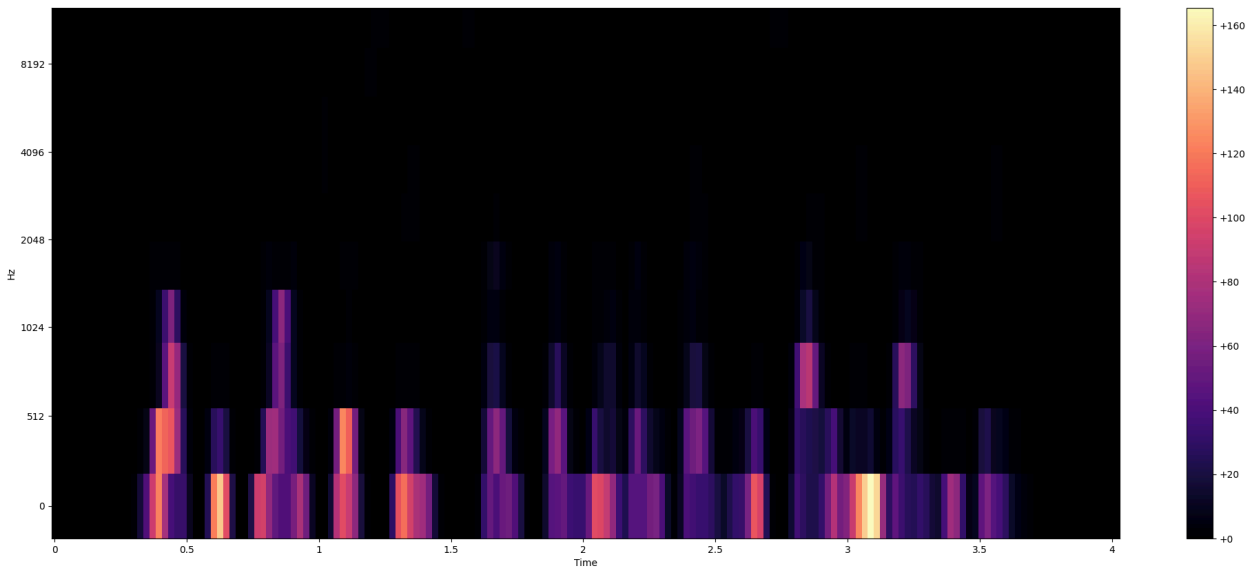
In [12]:

```
stft = librosa.stft(y)  # Short-time Fourier Transform
spectrogram = librosa.amplitude_to_db(np.abs(stft), ref=np.max)
plt.figure(figsize=(10,4))
librosa.display.specshow(spectrogram, sr=sr, x_axis='time', y_axis='linear')
plt.colorbar(format="%+2.f dB")
plt.title('Spectrogram (Hz)')
plt.xlabel('Time (s)')
plt.ylabel('Frequency (Hz)')
plt.show()
```

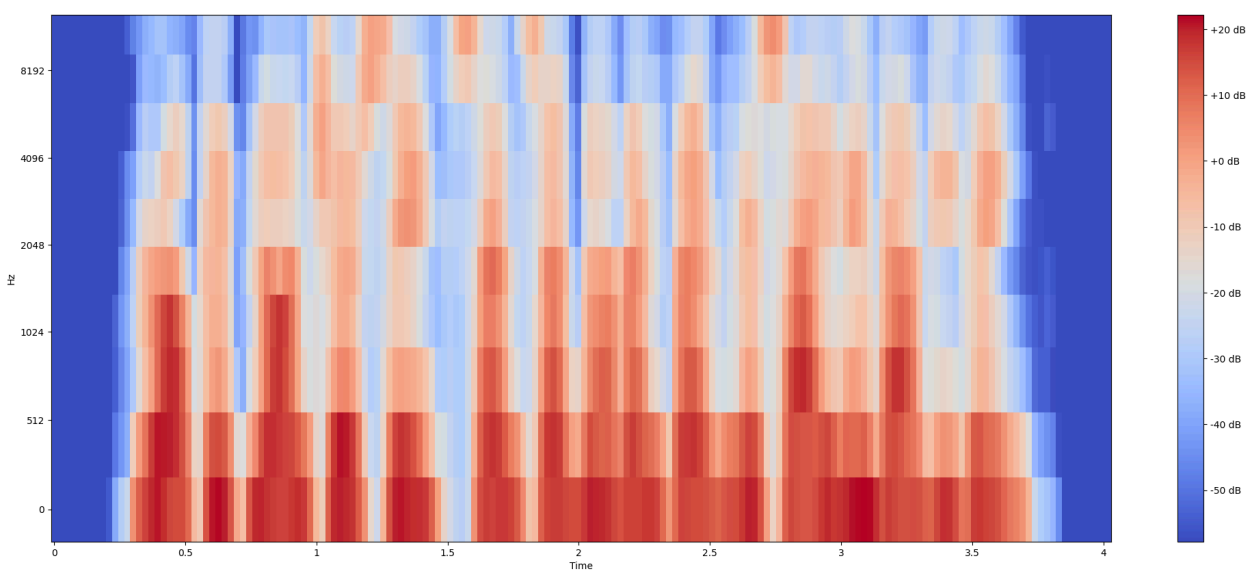


In [12]:

In [13]:
```
# Extracting the Mel spectrogram

mel_spectrogram = librosa.feature.melspectrogram(y=y, sr=sr, n_fft=2048, hop_l
plt.figure(figsize=(25, 10))
librosa.display.specshow(mel_spectrogram, x_axis="time", y_axis="mel", sr=sr)
plt.colorbar(format="%+2.f")
plt.show()
```

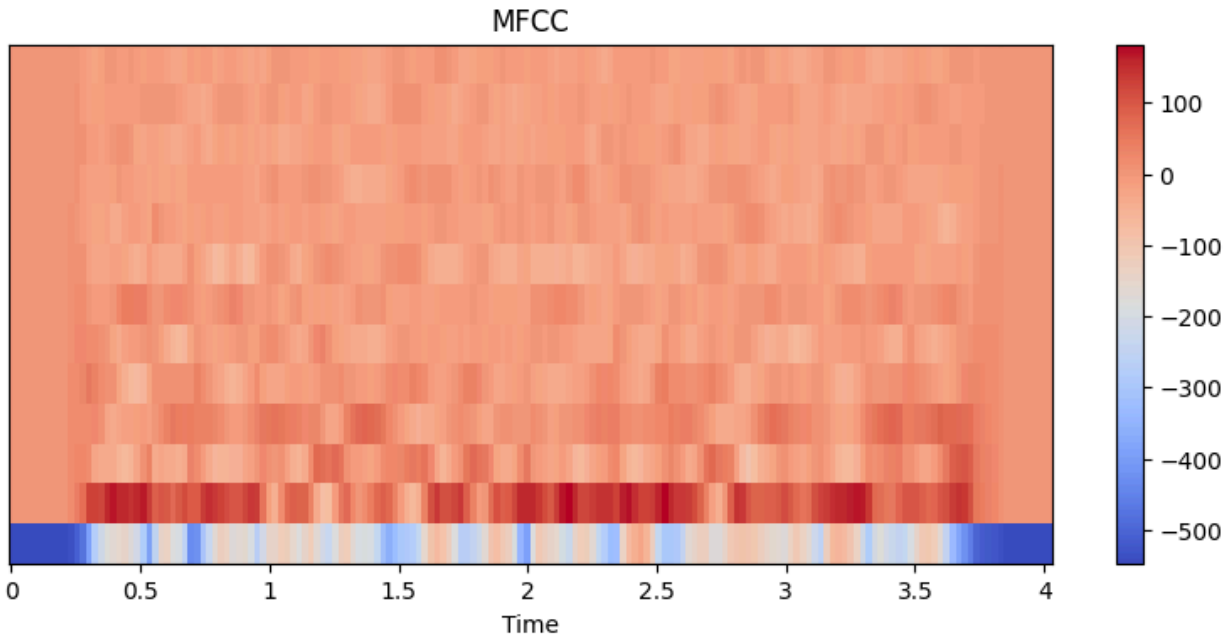


In [14]:
```
# Extracting the log mel spectrogram

log_mel_spectrogram = librosa.power_to_db(mel_spectrogram)
plt.figure(figsize=(25, 10))
librosa.display.specshow(log_mel_spectrogram, x_axis="time", y_axis="mel", sr=
plt.colorbar(format="%+2.f dB") # Added 'dB' to label for clarity
plt.show()
```



In [15]:
```
# Extracting MFCCs
mfcc = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=13)
```

```
plt.figure(figsize=(10, 4))
librosa.display.specshow(mfcc, x_axis='time', sr=sr)
plt.colorbar()
plt.title('MFCC')

plt.show()
```



In [16]:
```
# tabular form of mfcc
pd.DataFrame(mfcc)
```

Out[16]:

	0	1	2	3	4	5
0	-546.900452	-546.900452	-546.900452	-546.900452	-546.900452	-546.900452
1	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
2	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
3	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
4	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
5	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
6	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
7	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
8	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
9	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
10	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
11	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
12	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000

13 rows × 174 columns

In [17]:

```
# Extracting MFCCs for all files

def extract_features(directory):
  features_list = []

  audio_dir = glob(directory)

  for file in audio_dir:
    y, sr = librosa.load(file, sr = None)
    mfcc = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=13)
    mfcc_scaled = np.mean(mfcc.T, axis=0)
    duration = librosa.get_duration(y=y, sr=sr)
    data = {'filename': file, 'duration': duration}

    for i, val in enumerate(mfcc_scaled):
      data[f'mfcc_{i+1}'] = val
    features_list.append(data)
  return pd.DataFrame(features_list)
```

In [18]:

```
mfcc_df = extract_features("/content/drive/MyDrive/Hindi_fem_mono/Hindi_fem_au
```

In [19]:

```
mfcc_df
```

Out[19]:

	filename	duration	mfcc_1	mfcc_2	mfcc_3	mfcc_4
0	/content/drive/MyDrive/Hindi_fem_mono/Hindi_fe...	5.758333	-300.327148	120.928345	-8.470559	23.645855
1	/content/drive/MyDrive/Hindi_fem_mono/Hindi_fe...	5.169896	-319.762177	115.137024	-8.119379	32.418743
2	/content/drive/MyDrive/Hindi_fem_mono/Hindi_fe...	4.716750	-301.677856	114.540085	-8.549463	26.512072
3	/content/drive/MyDrive/Hindi_fem_mono/Hindi_fe...	5.005562	-295.948730	106.224686	-5.951683	32.565342
4	/content/drive/MyDrive/Hindi_fem_mono/Hindi_fe...	4.383125	-296.065399	120.751427	-12.603089	24.294432
...	...	...	...	...	...	...
4581	/content/drive/MyDrive/Hindi_fem_mono/Hindi_fe...	5.734375	-296.182678	114.870644	-15.997050	36.720207
4582	/content/drive/MyDrive/Hindi_fem_mono/Hindi_fe...	3.865021	-299.020538	109.225952	-3.182021	37.190197
4583	/content/drive/MyDrive/Hindi_fem_mono/Hindi_fe...	4.229604	-294.462524	94.881584	-16.818361	36.739426
4584	/content/drive/MyDrive/Hindi_fem_mono/Hindi_fe...	3.599417	-297.858032	89.722687	-15.981977	42.668259
4585	/content/drive/MyDrive/Hindi_fem_mono/Hindi_fe...	3.590187	-315.561981	121.084526	-2.824755	31.502188

4586 rows × 15 columns

Analysis

In [20]:
```
mfcc_df.drop(columns=['filename']).mean()
```

Out[20]:

	0
duration	5.349813
mfcc_1	-304.198914
mfcc_2	112.026276
mfcc_3	-11.472609
mfcc_4	27.720600
mfcc_5	4.816026
mfcc_6	2.630708
mfcc_7	-15.656408
mfcc_8	8.708613
mfcc_9	-3.109346
mfcc_10	-15.834547
mfcc_11	-2.584724
mfcc_12	-9.766752
mfcc_13	-8.544225

dtype: float64

In [21]:
```
mfcc_df.drop(columns=['filename']).std()
```

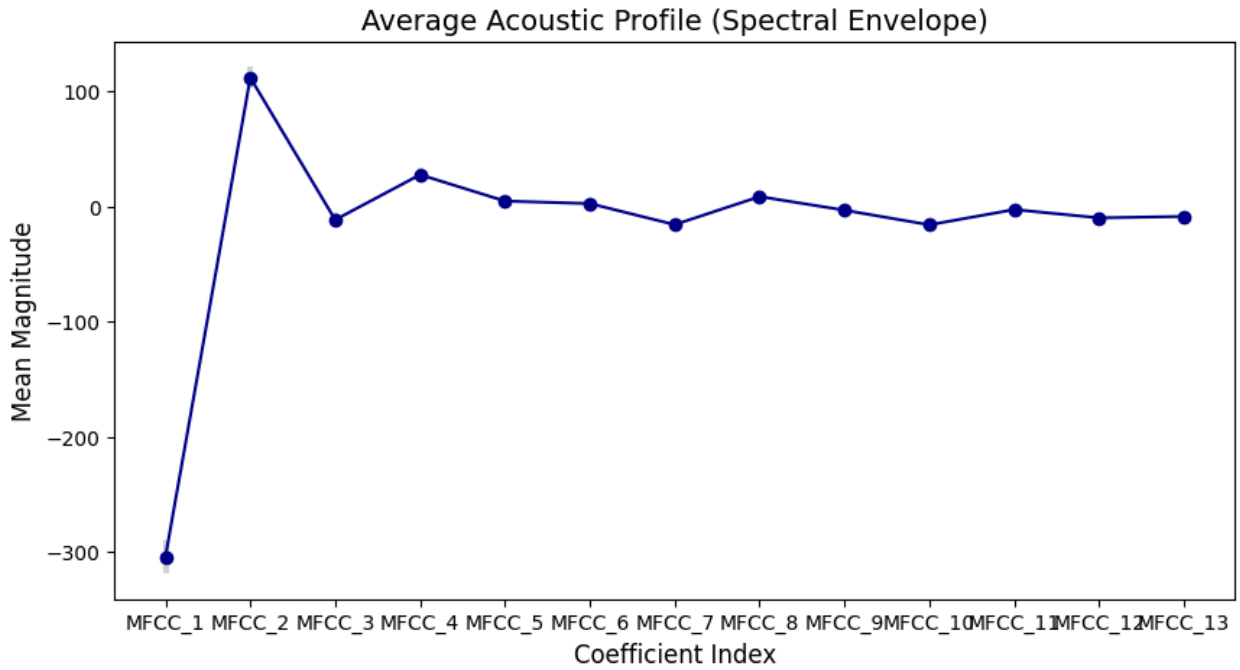
Out[21]:

	0
duration	1.233747
mfcc_1	14.629879
mfcc_2	9.457892
mfcc_3	5.875124
mfcc_4	6.478290
mfcc_5	4.950380
mfcc_6	3.816911
mfcc_7	3.817559
mfcc_8	3.614445
mfcc_9	2.825190
mfcc_10	3.448838
mfcc_11	3.591638
mfcc_12	2.419955
mfcc_13	2.517771

dtype: float64

In [22]:

```
# Calculate mean and std for each MFCC
means = mfcc_df.filter(like='mfcc').mean()
stds = mfcc_df.filter(like='mfcc').std()

plt.figure(figsize=(10, 5))
plt.errorbar(range(1, 14), means, yerr=stds, fmt='-o', color='darkblue',
             ecolor='lightgray', elinewidth=3, capsize=0)
plt.xticks(range(1, 14), [f'MFCC_{i}' for i in range(1, 14)])
plt.title('Average Acoustic Profile (Spectral Envelope)', fontsize=14)
plt.ylabel('Mean Magnitude', fontsize=12)
plt.xlabel('Coefficient Index', fontsize=12)
plt.show()
```


Average Acoustic Profile (Spectral Envelope)
Mean Magnitude
100
0
−100
−200
−300
MFCC_1
MFCC_2
MFCC_3
MFCC_4
MFCC_5
MFCC_6
MFCC_7
MFCC_8
MFCC_9
MFCC_10
MFCC_11
MFCC_12
MFCC_13
Coefficient Index